1 **(a)** A and D **[B1]**

(b) B and C **[B1]**

[Total Marks: 2]

2 **(a)** Continuous **[B1]**

(b) Discrete **[B1]**

(c) Continuous **[B1]**

[Total Marks: 3]

3 **(a)** **(i)** Discrete **[B1]**

(ii) Continuous **[B1]**

(iii) Continuous **[B1]**

(iv) Discrete **[B1]**

[Total Marks: 4]

4 **(a)** Discrete **[B1]**

(b) Continuous **[B1]**

(c) Continuous **[B1]**

(d) Continuous **[B1]**

[Total Marks: 4]

5 **(a)** Discrete **[B1]**

(b) Continuous **[B1]**

(c) Continuous **[B1]**

(d) Discrete **[B1]**

[Total Marks: 4]

6 **(a)** **(i)** Discrete **[B1]**

(ii) Continuous **[B1]**

(iii) Continuous **[B1]**

(iv) Discrete **[B1]**

(b) Width is measured (rather than counted) and can take any numerical value within a given interval **[B1]**

(c) A measurement of 6 metres would belong to both intervals (overlapping boundary) **[B1]**

$4 \leq w < 6$ (or $4 < w \leq 6$) **[B1]**

[Total Marks: 7]

7 (a) Lower bound = 37.5 **[B1]**

Upper bound = 42.5 **[B1]**

(b) Smallest possible number = 1248 **[B1]**

Largest possible number = 1252 **[B1]**

(Note: Continuous boundaries are 1247.5 and 1252.5, but as a discrete integer variable, it must take the integer values strictly within this range.)

[Total Marks: 4]

8 (a) Data Set A (time) is continuous, so it can take any decimal value and inequalities are needed to define exact boundary limits without gaps. **[B1]**

Data Set B (number of push-ups) is discrete, so it only takes integer values and a hyphen clearly defines the inclusive start and end integers. **[B1]**

(b) (i) 19.5 and 29.5 **[B1]**

(ii) 10 **[B1]**

(c) Age (usually given as an integer in years) OR Money/Currency (rounded to nearest cent/pence) OR Shoe size **[B1]**

[Total Marks: 5]

9 (a) k is discrete **[B1]**

m is continuous **[B1]**

(b) For one rod, $1.95 \leq r < 2.05$, and for the total mass, $19.95 \leq m < 20.05$ **[B1]**

$1.95k < 20.05$ giving $k < \dfrac{401}{39} = 10\dfrac{11}{39}$ **[M1]**

$2.05k > 19.95$ giving $k > \dfrac{399}{41} = 9\dfrac{30}{41}$ **[M1]**

Since k is discrete, k must be an integer, so $k = 10$ **[A1]**

(c) (i) Mean of $d = \dfrac{0(24) + 1(15) + 2(7) + 3(3) + 4(1)}{50} = \dfrac{42}{50}$ **[M1]**

0.84 **[A1]**

d is discrete with exact individual values known, whereas L is continuous data grouped into intervals so individual rod lengths are unknown and mid-interval values must be used **[B1]**

(ii) $\dfrac{18 + 17}{50} \times 100 = 70\%$ **[B1]**

L is continuous data grouped in the interval $50.0 \leq L < 50.2$, so the exact distribution of the 17 lengths within that interval is unknown **[B1]**

[Total Marks: 11]

10 **(a)** Continuous **[B1]**

(b) Height is continuous / can take decimal values. **[B1]**

A plant with a height between the intervals (e.g. 19.5 cm) would not fit into any class. **[B1]**

(c) Discrete **[B1]**

[Total Marks: 4]

11 **(a)** Discrete data can only take specific or exact values (often counted) **[B1]**

Continuous data can take any value within a given range (often measured) **[B1]**

(b) **(i)** Continuous **[B1]**

(ii) Rounding limits the precision of the recording, but the actual true length could still be any value between the rounded boundaries **[B1]**

[Total Marks: 4]

12 **(a)** Continuous **[B1]**

(b) Uses midpoints: 90, 110, 130, 150 **[M1]**

$\Sigma fx = 90 \times 12 + 110 \times 26 + 130 \times 15 + 150 \times 7 = 6940$ **[M1]**

$\frac{6940}{60} = 115.66... = 116$ or 115.7 **[A1]**

(c) The exact masses of the apples in each class are not known (or similar reference to data being grouped) **[B1]**

[Total Marks: 5]

13 **(a)** Length **[B1]**

(b) **(i)** 14.45 **[B1]**

(ii) 14.55 **[B1]**

(c) Identifies that the discrete values include $5, 6, 7, 8, 9$ OR uses continuous boundaries $9.5 - 4.5$ **[M1]**

5 **[A1]**

[Total Marks: 5]

14 **(a)** Continuous **[B1]**

(b) Identifies midpoints: $150, 250, 350, 450$ **[M1]**

15 **(a)** N is discrete (integer values only), so there are no values between 12 and 13 and inclusive inequalities ($10 \leq N \leq 12$, $13 \leq N \leq 15$) cover all counts without gaps **[B1]**

T is continuous, so strict upper inequalities ($48 \leq T < 50$, $50 \leq T < 52$) are needed to include all decimal values up to 50 without overlapping **[B1]**

(b) For N: $\sum fm = 6(11) + 14(14) + 12(17) + 8(20) = 626$ **[M1]**

Estimated mean of $N = \dfrac{626}{40} = 15.65$ **[A1]**

For T: $\sum fm = 5(49) + 15(51) + 13(53) + 7(55) = 2084$ **[M1]**

Estimated mean of $T = \dfrac{2084}{40} = 52.1$ seconds **[A1]**

(c) (i) The mean is a measure of average and does not have to be a possible value of a discrete variable **[B1]**

No, the student is incorrect; 15.65 is a valid mean even though N takes only integer values **[B1]**

(ii) Diagram for N: Bar chart (or vertical line graph) **[B1]**

Diagram for T: Histogram (or cumulative frequency curve / frequency polygon) **[B1]**

[Total Marks: 10]

16 (a) (i) Discrete **[B1]**

(ii) Continuous **[B1]**

(b) The variable can still only take specific, fixed intervals/values (e.g., jumping by 0.5) **[B1]**

It cannot take any value within a range (e.g., a shoe size of 8.23 does not exist) **[B1]**

[Total Marks: 4]

17 (a) Continuous data can take any value within a given range (measured). **[B1]**

Discrete data can only take specific, exact values (usually counted). **[B1]**

(b) (i) Continuous **[B1]**

(ii) Shoe size can only take specific, set values (there are gaps between sizes like 7 and 7.5). **[B1]**

It cannot take every possible value (e.g. you cannot have a shoe size of 7.23), so it must be discrete. **[B1]**

[Total Marks: 5]

18 (a) (i) Discrete **[B1]**

(ii) Continuous **[B1]**

(b) Upper bound of mass of one book = 855 g **[M1]**

$24 \times 855 = 20520$ g $\implies$ 20.52 **[A1]**

(c) Lower bound of empty box = 1.15 kg **[B1]**

Lower bound of mass of one book = 845 g = 0.845 kg seen **[B1]**

Total lower bound $= 1.15 + (24 \times 0.845) = 21.43$ **[A1]**

[Total Marks: 7]